



**MEPCO SCHLENK ENGINEERING COLLEGE, SIVAKASI (Autonomous)**  
**Department of Computer Science and Engineering**

**ASSIGNMENT – 2**

**Sub. Code & Name: 19CS603 – OBJECT ORIENTED ANALYSIS AND DESIGN**

Name of the Faculty: **Dr. P.Vinoth**

Class: **III Year B.E. CSE – 'B' Sec.**

Date of Announcement: **25.04.2025**

Date of Submission: **30.04.2025**

**Consider the following problem description for a Police Information System:**

The Police Information System helps officers track their assigned tasks. Officers may be assigned to investigate specific cases, patrol designated areas, or attend events such as accidents, court cases, and political meetings. Some assignments are ongoing, while others are limited to a specific period. The system's information is managed by the logistic administrator, while individual officers have an interface to view their assigned tasks.

| CL | #  | Questions                                                                                                                                                                                                                                                               | Mark | CO No. | PI                         |
|----|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|--------|----------------------------|
| A  | 1. | Draw the <b>State Machine Diagram</b> for the above-mentioned scenarios with the most suitable description.                                                                                                                                                             | (18) | CO4    | 4.2.1                      |
| A  | 2. | Draw the <b>Component and Deployment Diagram</b> for the above-mentioned scenarios with the most suitable description.                                                                                                                                                  | (16) | CO4    | 3.3.1<br>11.3.1            |
| U  | 3. | Describe in detail the following GRASP patterns with the problem statement, solution, most suitable diagram, contraindications, benefits, and related patterns.<br><br><b>A) Creator</b><br><b>B) Information Expert</b><br><b>C) High Cohesion &amp; Low Coupling.</b> | (16) | CO5    | 13.1.2<br>11.2.1<br>11.3.1 |

*Restriction: Make sure that all your assignment work should be unique and original.*